

Astrophysicist using SkyBrowser in a presentation about James Webb Space Telescope at the American Museum of Natural History.



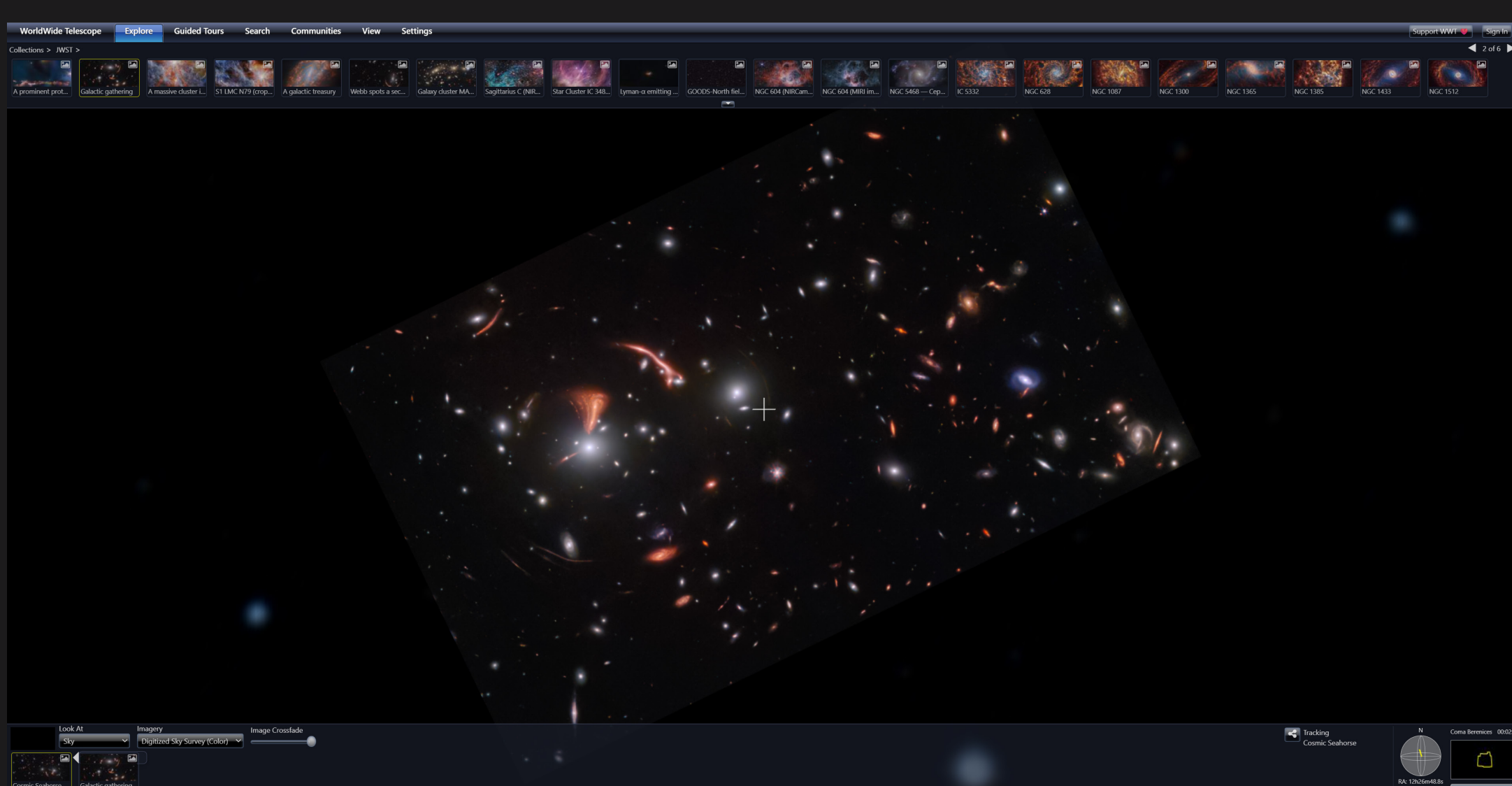
SkyBrowser: In-situ Dissemination of Deep Space Images

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Goals

- **Unify 2D and 3D astronomy** by integrating WorldWide Telescope (observations) with OpenSpace (immersive environments) to link spacecraft context and imagery.
- **Improve understanding and engagement** by enabling researchers and the general public to intuitively explore observations within their spatial and operational contexts in an interactive environment.

WorldWide Telescope



Screenshot of the user interface of WorldWide Telescope. The center displays a high-resolution telescope image layered over an all-sky survey.

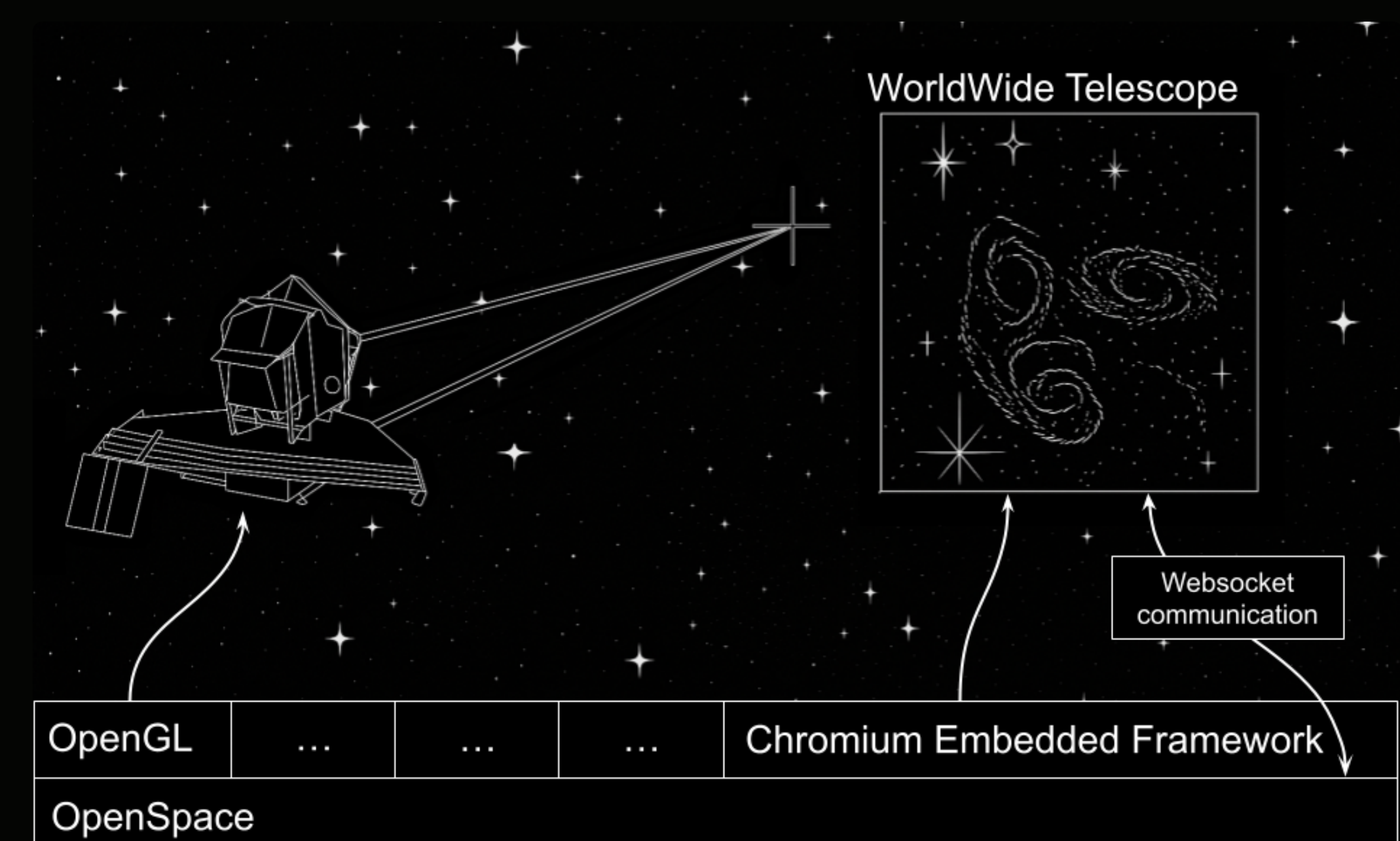
OpenSpace



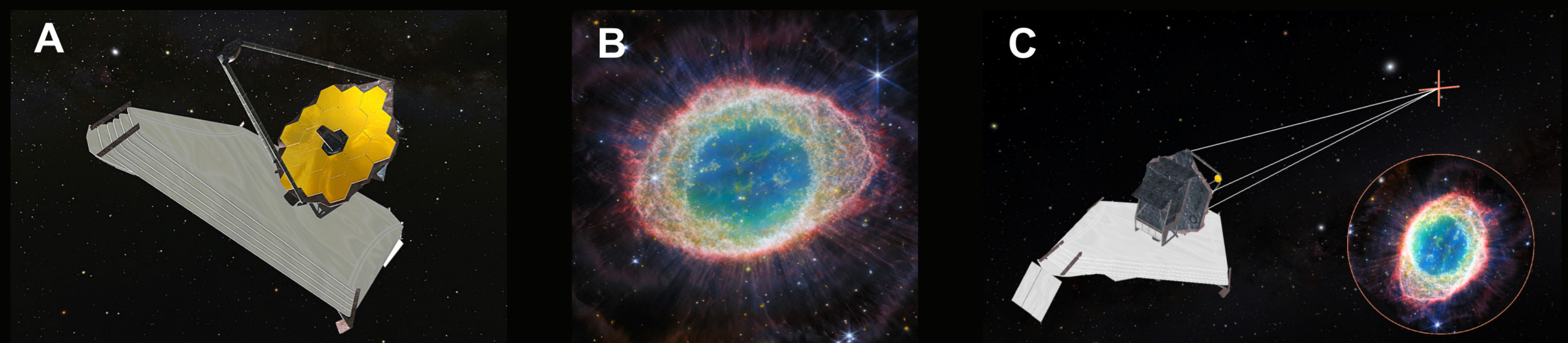
Screenshot of WorldWide Telescope integrated with OpenSpace, showing telescope imagery aligned with spacecraft position and field of view. The spacecraft position and field of view updates correctly based on the selected in-game time.

System

- **Architecture** WWT runs as a WebGL app inside Chromium Embedded Framework, connected to OpenSpace via WebSocket.
- **Coordinate conversion** Translates WWT's spherical RA/Dec into OpenSpace's Cartesian galactic coordinates, with added roll handling.
- **API extensions** Developed in collaboration with WWT team to support interactive control and image integration.
- **Level-of-Detail Rendering** Delivers interactive, tiled images of telescope data and all-sky surveys.



Overview of the system architecture.



Different visualization modalities for astronomy images. (A) A physically correctly animated 3D model that provides context for the location and orientation of the telescope. (B) High-resolution multi-wavelength images of astronomical objects that might be taken from different telescopes. (C) Embedding the 2D astronomy images into a 3D contextual environment to provide improved context for the location of astronomical images.

Use Case

- **Planetarium outreach** In daily use by dozens of planetariums worldwide to present space missions and telescope observations with up-to-date WWT imagery.
- **Timely discoveries** Shared databases enable researchers to immediately present to the public about new data.
- **Comparative insight** Shows the positions, fields of view and instrument scales for many spacecraft simultaneously, making mission capabilities easier to understand and compare.

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<https://github.com/OpenSpace>

